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EXPERIMENT – 2

Encryption – Crypto 101 using TryHackMe Platform

Note: Output is attached as screenshot at output section in this document

Aim

To understand the fundamentals of **cryptography** by performing hands-on exercises in the **Crypto 101** room on the **TryHackMe** platform, including basic encryption, encoding, hashing, and cryptographic concepts.

Tools / Platform Required

- Web browser (Chrome / Firefox)
- Internet connection
- TryHackMe account (free)
- Built-in TryHackMe terminal (AttackBox) or local Linux machine

Theory (Brief)

Cryptography is the practice of securing information by converting plain text into unreadable format (cipher text) using encryption techniques.

Crypto 101 introduces basic concepts such as:

- Encoding vs Encryption
- Symmetric and Asymmetric encryption
- Hashing
- Common ciphers (Caesar, Base64, etc.)

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Crypto 101 Room

1. In the search bar, type **Crypto 101**
2. Click on the room named **Crypto 101**
3. Click **Join Room**

Step 3: Start the Machine (if required)

1. Click **Start Machine / Start AttackBox**
2. Wait for the virtual environment to load.

Step 4: Perform Cryptography Tasks

Complete the tasks provided in the room, such as:

4.1 Encoding & Decoding

(a) Base64 Encoding / Decoding

Purpose:

Base64 is an **encoding technique** used to represent binary data in readable ASCII format.

How to Solve

To Decode Base64

1. Copy the given Base64 string from the question.

2. Open the TryHackMe terminal.
3. Use the command:

```
echo "Q3J5cHRvZ3JhcGh5" | base64 -d
```

4. The output will be the decoded plain text.
5. Enter the decoded text into the answer box.

To Encode Base64

```
echo "cryptography" | base64
```

Expected Output Example: Cryptography

(b) Hexadecimal Conversion

Purpose:

Hexadecimal is a base-16 encoding system commonly used in computing.

How to Solve

Hex to Text

```
echo 48656c6c6f | xxd -r -p
```

Text to Hex

```
echo "Hello" | xxd -p
```

Expected Output: Hello

4.2 Classical Ciphers

(a) Caesar Cipher

Purpose:

Caesar cipher shifts letters by a fixed number.

How to Solve

1. Identify the shift value (given or by trial).
2. Use online Caesar decoder or terminal tool.

Using terminal

```
echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
```

(Shift = 3)

Expected Output: HELLO

(b) Shift-Based Encryption

Purpose:

Similar to Caesar cipher but may use different shift values.

How to Solve

1. Try different shift values until meaningful text appears.
2. Online tools can be used if allowed.

Example:

- Shift 1 → incorrect
- Shift 3 → readable text → correct answer

4.3 Hashing

(a) Understanding Hash Algorithms

Hash Type	Length
MD5	32 characters
SHA-1	40 characters
SHA-256	64 characters

(b) Identifying Hashes

How to Solve

1. Count the number of characters in the given hash.
2. Match it with the table above.

Example:

5d41402abc4b2a76b9719d911017c592

→ 32 characters → MD5

Generating Hashes

```
echo -n "hello" | md5sum
```

```
echo -n "hello" | sha1sum
```

```
echo -n "hello" | sha256sum
```

4.4 Encryption Concepts

(a) Encryption vs Hashing

Encryption

Reversible

Hashing

Irreversible

Uses keys

No keys

Used for secure communication

Used for password storage

(b) Symmetric vs Asymmetric Encryption

Symmetric

One key

Fast

Example: AES

Asymmetric

Two keys

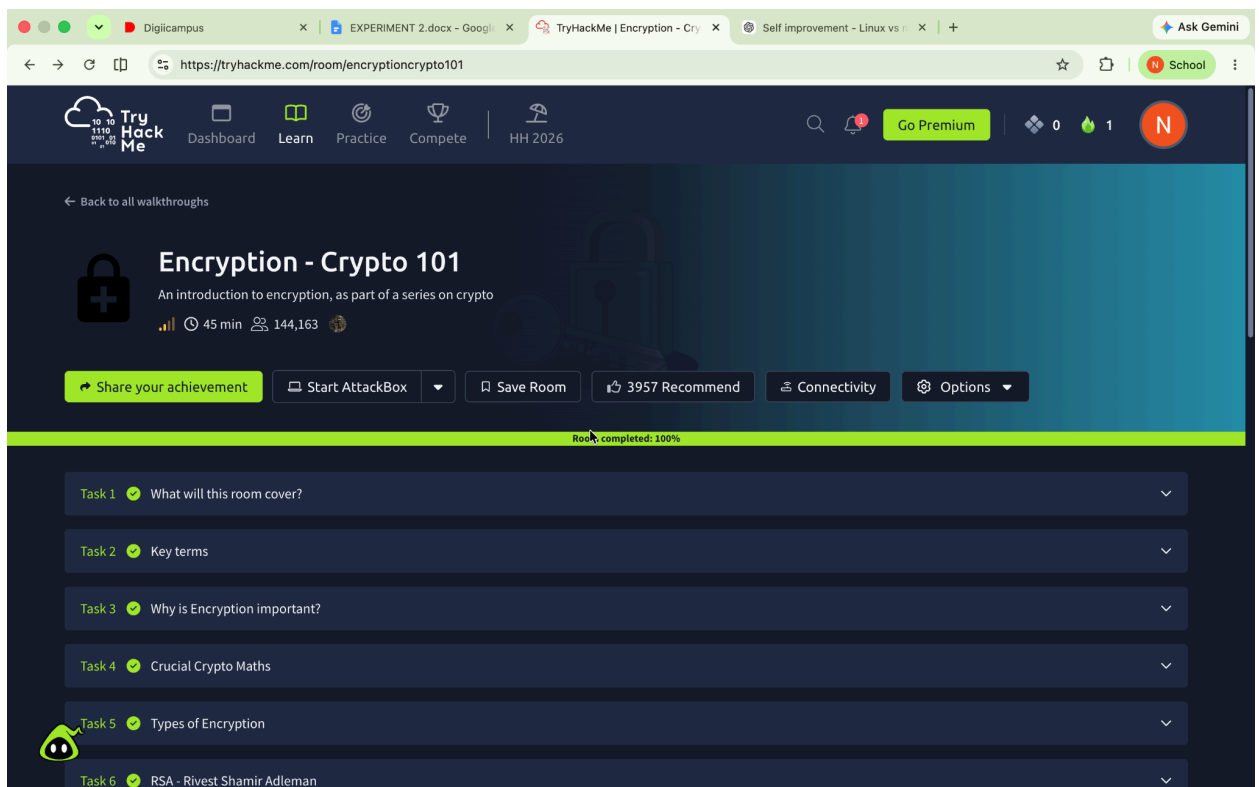
Slower

Example: RSA

Step 5: Verify Completion

1. Ensure all tasks are marked **Completed**
2. Observe the success message at the end of the room

Output



```
Applications Places System > Mon Aug 10, 15:58 AttackBox IP
root@ip-10-48-191-20: ~
File Edit View Search Terminal Help
root@ip-10-48-191-20:~# echo "Q3J5cHRvZ3JhcGh5" | base64 -d
Cryptographyroot@ip-10-48-191-20:~#
root@ip-10-48-191-20:~# echo "cryptography" | base64
Y3J5cHRvZ3JhcGh5Cg==
root@ip-10-48-191-20:~#
root@ip-10-48-191-20:~# echo 48656c6c6f | xxd -r -p
Helloroot@ip-10-48-191-20:~#
root@ip-10-48-191-20:~# echo "Hello" | xxd -p
48656c6c6f0a
root@ip-10-48-191-20:~#
root@ip-10-48-191-20:~# echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
HELLO
root@ip-10-48-191-20:~# echo -n "hello" | sha1sum
aaf4c61ddcc5e8a2dabede0f3b482cd9aea9434d -
root@ip-10-48-191-20:~#
root@ip-10-48-191-20:~# echo -n "hello" | sha256sum
2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824 -
root@ip-10-48-191-20:~#
```

Result

Thus, the **Crypto 101** experiment was successfully completed on the **TryHackMe platform**.

The experiment helped in understanding **basic cryptographic concepts**, encryption techniques, encoding methods, and hashing algorithms through hands-on practice.